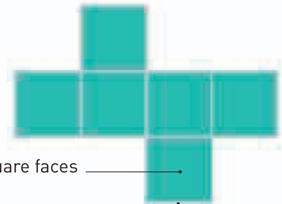
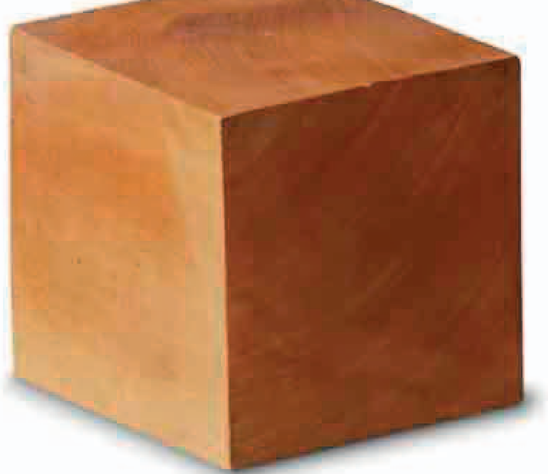
Platonic solids

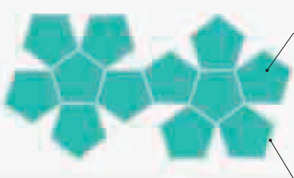
There are only five convex polyhedra (solids having several sides) that can be formed by joining identical polygons (shapes with three or more sides). Known as the Platonic solids, they are the cube (hexahedron), tetrahedron, octahedron, dodecahedron, and icosahedron.





Hexahedron (cube)

Dodecahedron

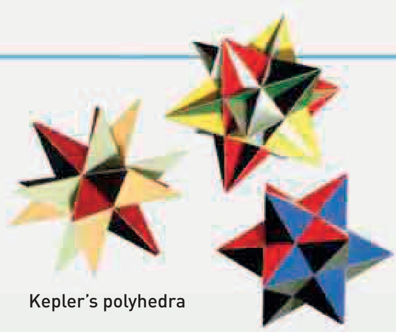


Icosahedron

SPHERICAL GEOMETRY

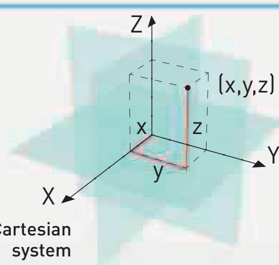
So-called "spherical geometry" allows the calculation of angles and areas on spherical surfaces, such as points on a map or the positions of stars and planets on the imaginary celestial sphere used by astronomers. This system does not follow all Euclidean rules. In spherical geometry, the three angles in a triangle sum to more than 180 degrees and parallel lines eventually intersect.

> " LET NO ONE **DESTITUTE OF GEOMETRY** COME UNDER MY ROOF. "
>
> **Plato, Greek philosopher and mathematician,** *c.*427–347 BCE



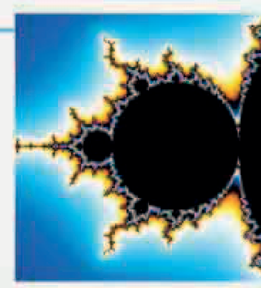
Kepler's polyhedra

1637
Analytic geometry
René Descartes's influential work *La Géometrie* introduces the idea that points in space can be measured with coordinate systems, and that geometrical structures can be described by equations—a field known as analytic geometry.



Cartesian system

20th century
Fractal geometry
Computing power allows fractals—equations in which detailed patterns repeat on varying scales—to be illustrated in graphical form, producing iconic images such as the famous Mandelbrot set.



Mandelbrot fractal

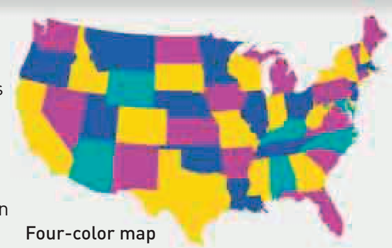
Möbius strip

1858
Topology
Mathematicians become fascinated by topology—edges and surfaces, rather than specific shapes. The iconic Möbius strip is an object with a single surface and a single continuous edge.

1882
Klein discovery
Investigating geometries with more than three dimensions, German scholar Felix Klein discovers a construct with no surface boundaries.

Modern Klein bottle

Present day
Computerized proofs
Computer power solves problems such as the four-color theorem (only four colors are needed to distinguish between regions of even complex maps).



Four-color map